A simplified model having additive noises

- Until now, we have assumed that our system model can have a very general nonlinear state-space model form:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, w_k) \\ z_k &= h(x_k, u_k, v_k). \end{aligned}$$

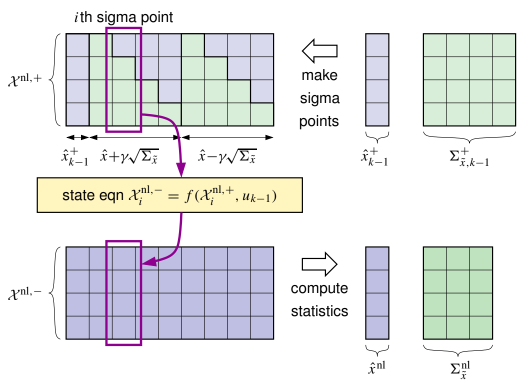
- Since the SPKF must jointly model all random variables in the nonlinear dynamics, we needed to create $\hat{x}_k^a = \left[\hat{x}_k^T, w_k^T, v_k^T\right]^T$ and corresponding $\Sigma_{\tilde{x},k}^a$.
- But, if the noises add linearly to the state and output equations, we can avoid the complexity of needing to create augmented state estimate and covariance.
- Here we consider that case, where we now assume the model form:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k) + w_k \\ z_k &= h(x_k, u_k) + v_k. \end{aligned}$$

Step 1a

- We seek to compute the predicted state:

$$\hat{x}_k^- = \mathbb{E}[x_k \mid \mathbb{Z}_{k-1}] = \underbrace{\mathbb{E}[f(x_{k-1}, u_{k-1}) | \mathbb{Z}_{k-1}]}_{\hat{x}_k^{\text{nl}}} + \underbrace{\mathbb{E}[w_{k-1} \mid \mathbb{Z}_{k-1}]}_{\bar{w}_{k-1}}.$$



- We use a sigma-point approach to approximate the nonlinear first term.
- Compute state-prediction sigma points: $\mathcal{X}_i^{\text{nl},-} = f(\mathcal{X}_i^{\text{nl},+}, u_{k-1})$.
- Compute the mean and covariance of the nonlinear first term:

$$\hat{x}^{\text{nl}} = \sum_{i=0}^{p} \alpha_i^{(\text{m})} \mathcal{X}_i^{\text{nl},-}$$

$$\Sigma_{\tilde{x}}^{\text{nl}} = \sum_{i=0}^{p} \alpha_i^{(\text{c})} (\mathcal{X}_i^{\text{nl},-} - \hat{x}^{\text{nl}})(\mathcal{X}_i^{\text{nl},-} - \hat{x}^{\text{nl}})^T.$$

Step 1b

- Combining these results, we have: $\hat{x}_k^- = \hat{x}^{\text{nl}} + \bar{w}_{k-1}$.
- Next, to find the covariance for Step 1b.

$$\begin{aligned} \Sigma_{\tilde{x},k}^- &= \mathbb{E}[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T] \\ &= \mathbb{E}\Big[\big(f(x_{k-1}, u_{k-1}) + w_{k-1} - \hat{x}^{\text{nl}} - \bar{w}_{k-1}\big)\big(f(x_{k-1}, u_{k-1}) + w_{k-1} - \hat{x}^{\text{nl}} - \bar{w}_{k-1}\big)^T\Big] \\ &= \mathbb{E}\Big[\big(f(x_{k-1}, u_{k-1}) - \hat{x}^{\text{nl}}\big)\big(f(x_{k-1}, u_{k-1}) - \hat{x}^{\text{nl}}\big)^T + (w_{k-1} - \bar{w}_{k-1})(w_{k-1} - \bar{w}_{k-1})^T \\ &\quad + \big(f(x_{k-1}, u_{k-1}) - \hat{x}^{\text{nl}}\big)(w_{k-1} - \bar{w}_{k-1})^T + (w_{k-1} - \bar{w}_{k-1})\big(f(x_{k-1}, u_{k-1}) - \hat{x}^{\text{nl}}\big)^T\Big]. \end{aligned}$$

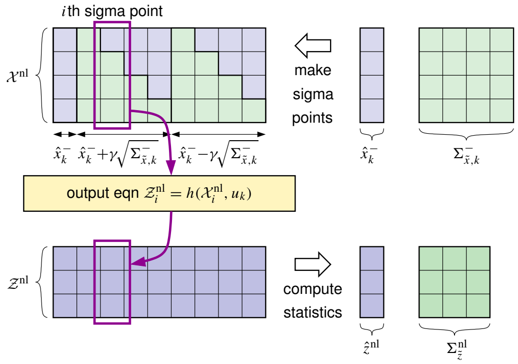
- If we assume that the state is uncorrelated with present and future process noise:

$$\begin{aligned} \Sigma_{\tilde{x},k}^- &= \mathbb{E}\left[\big(f(x_{k-1}, u_{k-1}) - \hat{x}^{\text{nl}}\big)\big(f(x_{k-1}, u_{k-1}) - \hat{x}^{\text{nl}}\big)^T\right] \\ &\quad + \mathbb{E}\left[(w_{k-1} - \bar{w}_{k-1})(w_{k-1} - \bar{w}_{k-1})^T\right] = \Sigma_{\tilde{x}}^{\text{nl}} + \Sigma_{\widetilde{w}}. \end{aligned}$$

Step 1c

- Now, to compute the predicted measurement:

$$\hat{z}_k = \mathbb{E}[z_k \mid \mathbb{Z}_{k-1}] = \underbrace{\mathbb{E}[h(x_k,u_k) \mid \mathbb{Z}_{k-1}]}_{\hat{z}_k^{\text{nl}}} + \underbrace{\mathbb{E}[v_k \mid \mathbb{Z}_{k-1}]}_{\bar{v}_{k-1}}.$$



- Existing sigma points $\mathcal{X}_k^{\text{nl},-}$ do not contain $\bar{w}_{k-1}$ or $\Sigma_{\tilde{w}}$ components.
- Need to generate new sigma points $\mathcal{X}^{\text{nl}}$ to represent $\hat{x}_k^-$. Then, $\mathcal{Z}_i^{\text{nl}} = h(\mathcal{X}_i^{\text{nl}}, u_k)$.
- Compute the mean and covariance of the nonlinear first term:

$$\hat{z}^{\text{nl}} = \sum_{i=0}^{p} \alpha_i^{(\text{m})} \mathcal{Z}_i^{\text{nl}}$$

$$\Sigma_{\tilde{z}}^{\text{nl}} = \sum_{i=0}^{p} \alpha_i^{(\text{c})} (\mathcal{Z}_i^{\text{nl}} - \hat{z}^{\text{nl}})(\mathcal{Z}_i^{\text{nl}} - \hat{z}^{\text{nl}})^T.$$

Step 1c (continued) through 2a

- Combining these results, we have: $\hat{z}_k = \hat{z}^{\text{nl}} + \bar{v}_k$.
- For Step 2a, we need to find $\Sigma_{\tilde{z},k}$ and $\Sigma_{\tilde{x}\tilde{z},k}^-$:

$$\begin{aligned}
\Sigma_{\tilde{z},k} &= \mathbb{E}[(z_k - \hat{z}_k)(z_k - \hat{z}_k)^T] \\
&= \mathbb{E}\Big[\big(h(x_k,u_k) + v_k - \hat{z}^{\text{nl}} - \bar{v}_k\big)\big(h(x_k,u_k) + v_k - \hat{z}^{\text{nl}} - \bar{v}_k\big)^T\Big] \\
&= \mathbb{E}\Big[\big(h(x_k,u_k) - \hat{z}^{\text{nl}}\big)\big(h(x_k,u_k) - \hat{z}^{\text{nl}}\big)^T + (v_k - \bar{v}_k)(v_k - \bar{v}_k)^T \\
&\qquad + \big(h(x_k,u_k) - \hat{z}^{\text{nl}}\big)(v_k - \bar{v}_k)^T + (v_k - \bar{v}_k)\big(h(x_k,u_k) - \hat{z}^{\text{nl}}\big)^T\Big].
\end{aligned}$$

- If we assume that the state is uncorrelated with present measurement noise:

$$\begin{aligned}
\Sigma_{\tilde{z},k} &= \mathbb{E}\Big[\big(h(x_k,u_k) - \hat{z}^{\text{nl}}\big)\big(h(x_k,u_k) - \hat{z}^{\text{nl}}\big)^T\Big] + \mathbb{E}\Big[(v_k - \bar{v}_k)(v_k - \bar{v}_k)^T\Big] \\
&= \Sigma_{\tilde{z}}^{\text{nl}} + \Sigma_{\tilde{v}}.
\end{aligned}$$

Step 2a (continued) through 2c

- We compute $\Sigma_{\tilde{x}\tilde{z},k}$ using a sigma-point approach:

$$\Sigma_{\tilde{x}\tilde{z},k}^- = \sum_{i=0}^{p} \alpha_i^{(\text{c})} (\mathcal{X}_i^{\text{nl}} - \hat{x}_k^-)(\mathcal{Z}_i^{\text{nl}} - \hat{z}^{\text{nl}}).$$

- Then, the remainder of Steps 2a through 2c are standard:

$$\begin{aligned}
L_k &= \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + L_k(z_k - \hat{z}_k) \\
\Sigma_{\tilde{x},k}^+ &= \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T.
\end{aligned}$$

Summary

- When noises are additive, we can implement an SPKF without augmenting the state vector before finding sigma points.
- But, now two Cholesky decompositions are needed. Is the new method faster?
  - It depends on the dimensions of w_k and v_k, speed of state and output equations.
- One choice of tuning for this method produces the "cubature Kalman filter" (CKF):

$$\gamma = \sqrt{n_x}, \quad \text{and} \quad \alpha_i^{(\mathrm{m})} = \alpha_i^{(\mathrm{c})} = \begin{cases} 0, & i = 0 \\ 1/(2n_x), & i \neq 0. \end{cases}$$

  - Eliminates user-defined tuning parameters and negative α; one fewer sigma point.
- Can approximate the new method by generating $\mathcal{X}_i^{\mathrm{nl}} \approx \mathcal{X}_i^{\mathrm{nl},-} + \bar{w}_{k-1}$.
  - This avoids a second Cholesky decomposition but ignores the randomness of $\Sigma_{\widetilde{w}}$ when generating $\mathcal{X}_i^{\mathrm{nl}}$, which will bias the estimate $\hat{z}_k$ and its covariance.
  - It won't be as accurate, but it might be "good enough."